Step Growth (Condensation) polymerization

A condensation takes place between two polyfunctional molecules to produce one larger polyfunctional molecule with the *possible* elimination of a small molecule such as H_2O, NH_3, HCl, etc. The reaction continues until one of the reagents is used up.

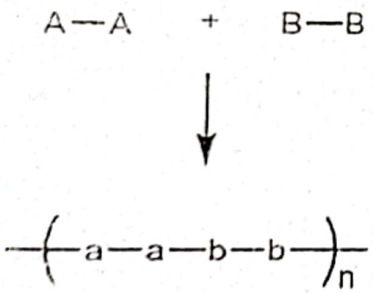
Common types:

Polyamides
Polyesters
Polycarbonates
Polyurethanes

CamScanner

Step Growth Polymerizations

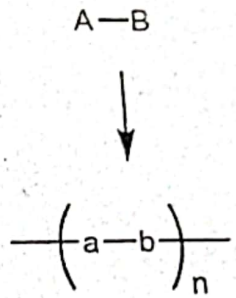
Bifunctional monomers



A-A + B-B ⟶ A-AB-B ⟶ B-BA-AB-BA-AB-BA-AB-B

1-mer 1-mer 2-mer poly-mer

AB monomers (Self condensation)



A-B A-BA-B $\{A\text{-}B\}_n$

1-mer 2-mer poly-mer

1. Difunctional monomers ⟹ Linear polymers

   AB type, AA and BB

2. Polyfunctional monomers ⟹ Network polymers

   Functionality > 2

3. Polymers retain their functionality as end groups at the completion of polymerization

4. Single reaction (cf. initiation, propagation,and termination in chain-reaction polymerization)

5. M.W. increases slowly

   Carothers equation

$$\overline{DP} = \frac{1}{1-p}$$

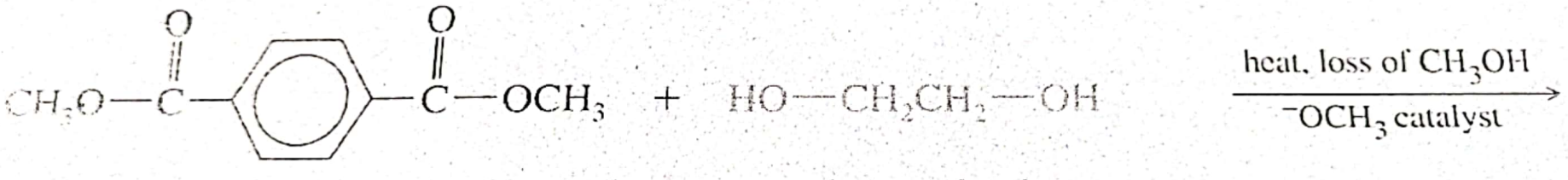
p = extent of reaction

6. High-yield reactions and an exact stoichiometric balance are necessary to obtain high-m-w linear polymer

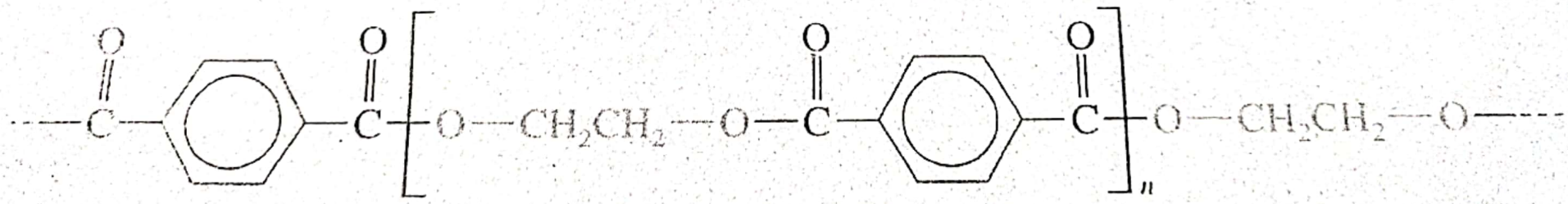
CamScanner

Polyesters

- Polymer of terephthalic acid and ethylene glycol.
- Made by the transesterification of the methyl ester.



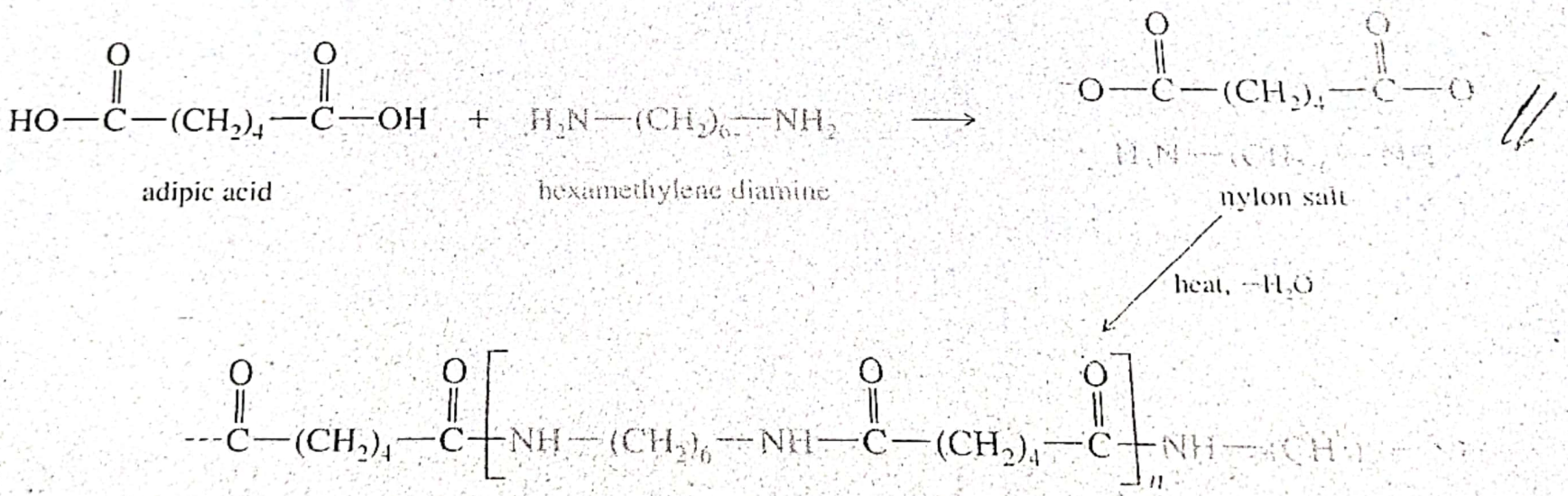
dimethyl terephthalate + ethylene glycol



poly(ethylene terephthalate) or PET, also called Dacron® polyester or Mylar® film

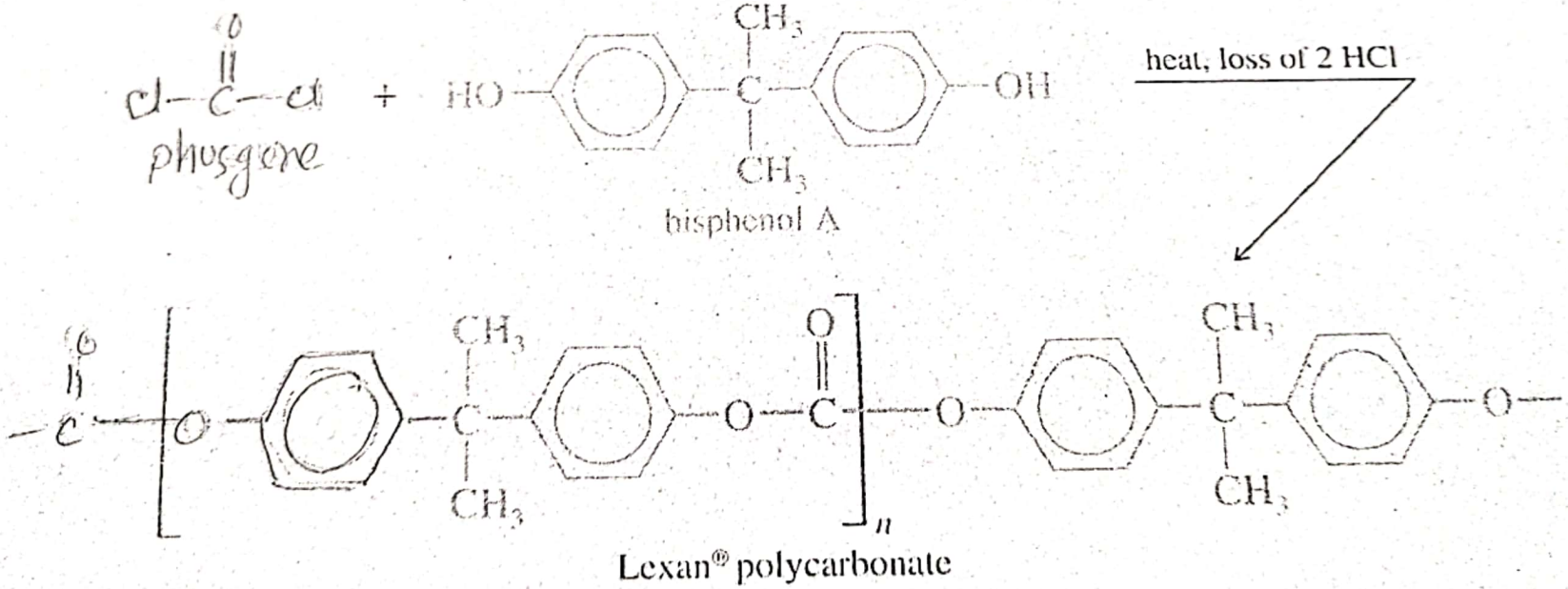
Polyamides: Nylon

Usually made from reaction of diacids with diamines, but may also be made from a single monomer with an amino group at one end and acid group at other.



Polycarbonates

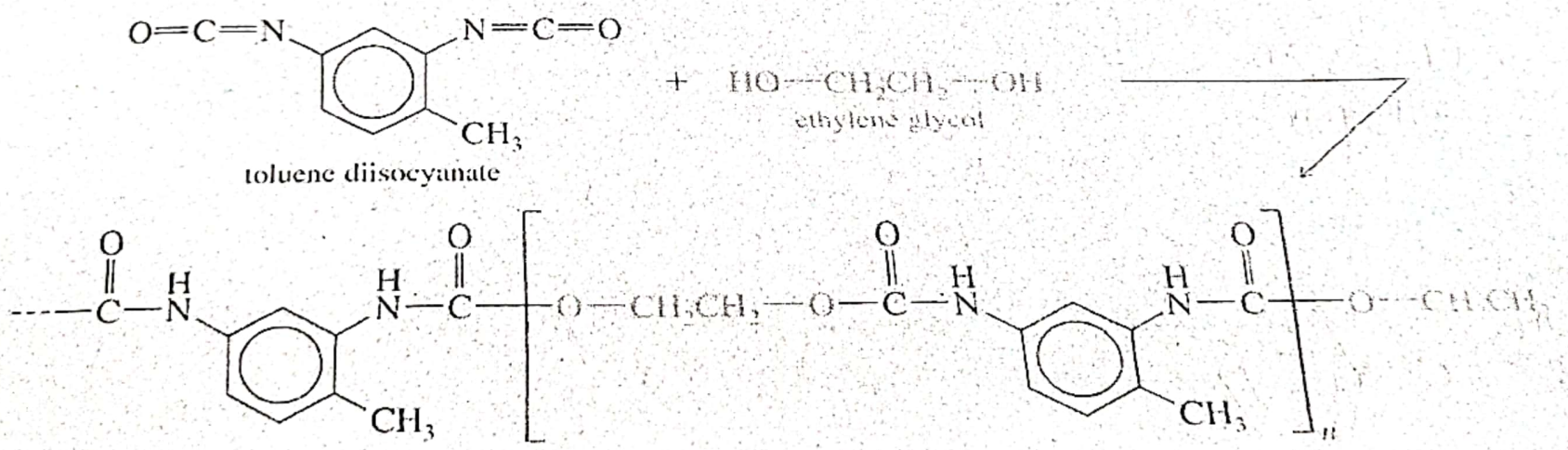
- Esters of carbonic acid.
- React phosgene with bisphenol A to obtain Lexan for bulletproof windows.



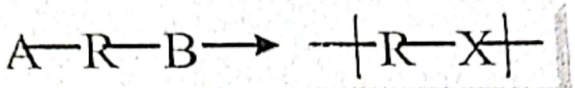
CamScanner

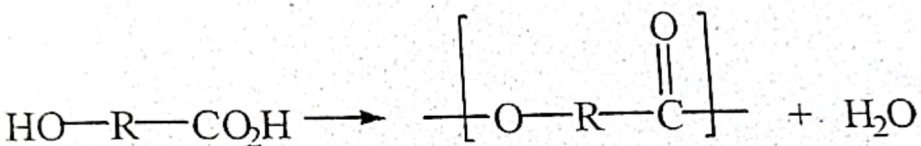
Polyurethanes

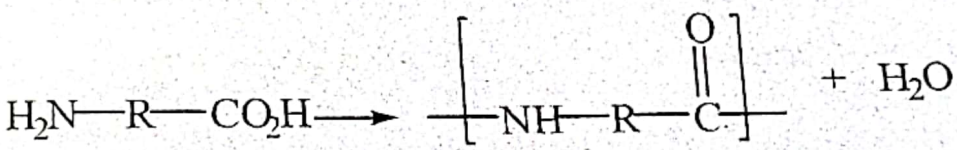
- Esters of carbamic acid, R-NH-COOH.
- Urethanes are prepared by reacting an alcohol with isocyanate.
- Polyurethanes are prepared by reacting a diol with a diisocyanate.



One having both reactive functional groups in one molecule







Monomer to have difunctional group

Self condensation

$$n\ HO-(CH_2)_5-\underset{\displaystyle O}{\overset{\displaystyle \|}{C}}-OH \longrightarrow \left[(CH_2)_5-\underset{\displaystyle O}{\overset{\displaystyle \|}{C}}-O\right]_n + n\ H_2O$$

hydroxycaproic acid — polycaprolactone

MW in Step-Growth polymerization

Polymer MW in step-growth polymerization is determined by the fractional conversion, p, of the monomer during the polymerization.

Extent of reaction *p*:

$$p = \frac{\textit{number of functional groups that have reacted}}{\textit{number of functional groups originally present}}$$

One way to express molecular weight is the *degree of polymerization*- number of repeating units in the chain. Usually *average degree of polymerization* ($\overline{X_n}$ or $\overline{DP}$) is use as polymerization mechanism yields distribution of molecular weight.

CamScanner

Carothers equation

Consider a flask of monomer ....

If there are N_o molecules in the flask at time = 0 and N remaining at time t,

then the $\overline{DP}$ at time t is the average degree of polymerization... must just be, N_0/N.

The amount reacted is N_0-N and we can define p as the "conversion" or fraction reacted then as

$$p = (N_o - N) / N_o \text{ or}$$

$$N = N_o(1 - P)$$

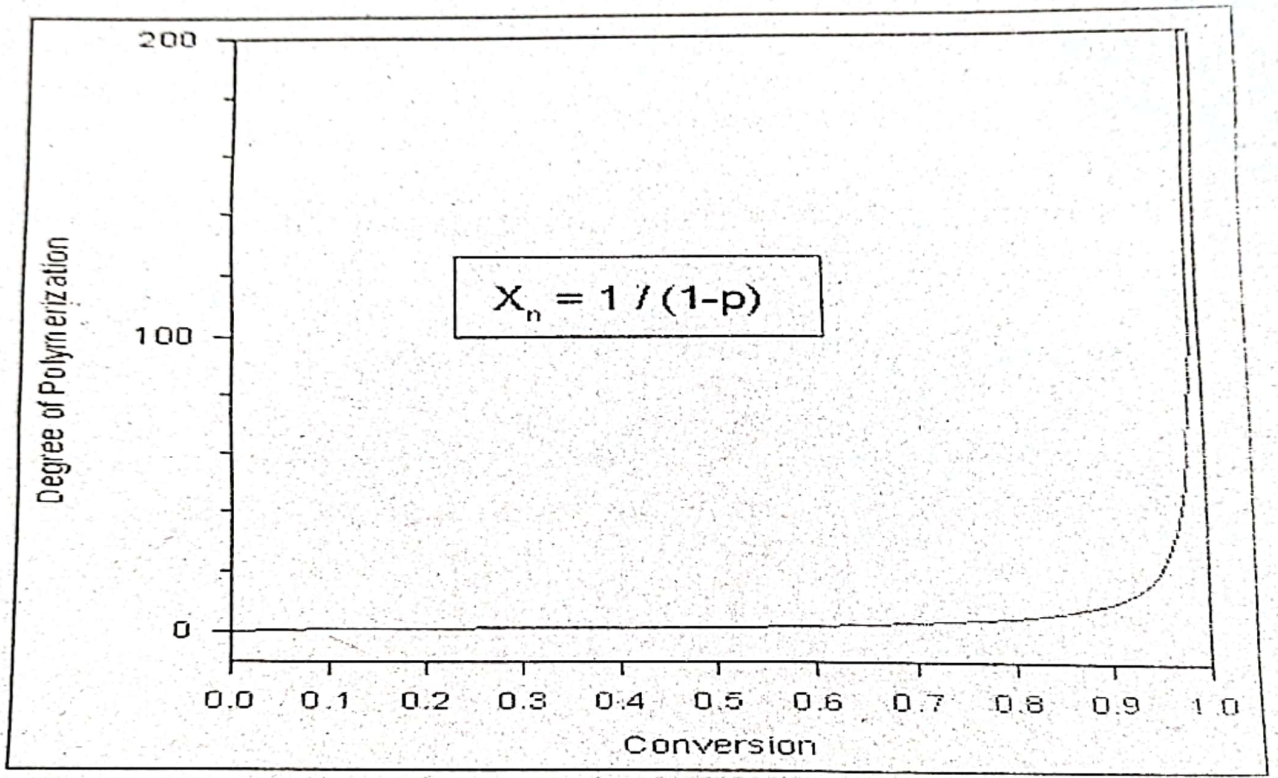
If $\overline{DP}$ is the average degree of polymerization... N_0/N ....substituting gives

$$\overline{DP} = 1 / (1 - P)$$

and for P = 0.98 (98% conversion), $\overline{DP}$ = only 50!

Carothers equation - relates/calculate molecular weight to the extent of a step growth polymerisation

CamScanner


200
100
0
Degree of Polymerization
$X_n = 1 / (1-p)$
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
Conversion

Number-average degree of polymerisation, $\overline{X}_n = N_0/N$

Conversion, $p = (N_0-N)/N_0 \longrightarrow N = N_0(1-p)$

Carothers equation

$$\overline{X}_n = 1/(1-p)$$

Number-average molecular weight: $\overline{M}_n = \overline{X}_n \times M_0 = M_0/(1-p)$
where M_0 is the molecular weight of the repeating unit

The weight-average degree of polymerisation, $\overline{X}_w = (1+p)/(1-p)$

Step – growth polymerization : MW distribution

Molecular weight distribution (polydispersity index) is $= \dfrac{\overline{X}_w}{\overline{X}_n} = \dfrac{(1+p)/(1-p)}{1/(1-p)} = (1+p)$

As $p \rightarrow 1$, MWD =2 for any linear step – growth polymerization .

CamScanner

KINETICS OF STEP GROWTH POLYMERIZATION

WHY BOTHER ?

- How long does it take to make polymer ?
- Can we speed up the reaction ?
- What is the relationship between kinetics and the Mol. Wt. Of the product ?

CamScanner

KINETICS OF POLYCONDENSATION

Let us consider the reaction below

$$A - A + B - B \longrightarrow A - AB - B$$

Consider a polyesterification of bifunctional monomers in which initially equal amounts of di-basic acid and glycol are present.

$$HOOC - R_1 - COOH + OH - R_2 - OH \longrightarrow \text{-----} - \overset{\overset{O}{\|}}{CO} \text{-----}$$

Case 1: Acid Catalyzed Polyesterification :

If the polyesterification is first order with respect to each functional group reactant, A and B, then the rate expression is

$$\text{Reaction Rate} = -\frac{d[A]}{dt} = k[A][B] \quad \text{--------(1)}$$

Given the fact that an exact stoichiometric balance is usually a necessary condition for achieving high molecular weight, we can equate [A] and [B] and the rate expression becomes,

$$-\frac{d[A]}{dt} = k[A]^2 \quad \text{----------} \quad (2)$$

or, by integration,

$$\frac{1}{[A]} - \frac{1}{[A_0]} = kt \quad \text{----------} \quad (3)$$

At any particular , t, $\overline{DP}$ is equal to the ratio of monomer molecules present initially to the total number at that time; that is

$$\overline{DP} = \frac{[A_0]}{[A]} \quad \text{----------} \quad (4)$$

Now the Carothers equation is

$$\overline{DP} = \frac{1}{1-p} \quad \text{----------} \quad (5)$$

CamScanner

Now combining eq. (4) and (5) and solving for [A],

$$[A] = [A_0](1-p) \text{ ---------- (6)}$$

By substituting this value into eq. (3), we obtain

$$\frac{1}{[A_0](1-p)} - \frac{1}{[A_0]} = kt \text{ ---------- (7)}$$

Since $\overline{DP} = 1/(1-p)$

$$\frac{\overline{DP}}{[A_0]} - \frac{1}{[A_0]} = kt \text{ ---------- (8)}$$

Or

$$\overline{DP} = [A_0]\,kt + 1$$

Case 2: No Acid Catalyst Added :

In the absence of added acid, the carboxylic acid assumes the role of catalyst and the rate expression becomes,

$$-\frac{d[A]}{dt} = k[A]\,[A][B]$$

That is

$$-\frac{d[A]}{dt} = k[A]^2[B] \quad \text{----------} \quad (1)$$

Again assuming [A] and [B] are equal,

$$-\frac{d[A]}{dt} = k[A]^3 \quad \text{----------} \quad (2)$$

Integration then gives

$$\frac{1}{[A]^2} - \frac{1}{[A_0]^2} = 2kt \quad \text{----------} \quad (3)$$

CamScanner

Substituting for [A] as above and rearranging

$$\frac{1}{[A_0]^2 (1-p)^2} - \frac{1}{[A_0]^2} = 2kt \quad \text{----------} \quad (4)$$

$$\frac{1}{(1-p)^2} = 2kt\,[A_0]^2 + 1 \quad \text{----------} \quad (5)$$

$$\overline{(DP)^2} = 2kt\,[A_0]^2 + 1$$

Substituting for [A] as above and rearranging

$$\frac{1}{[A_0]^2 (1-p)^2} - \frac{1}{[A_0]^2} = 2kt \quad \text{----------} \quad (4)$$

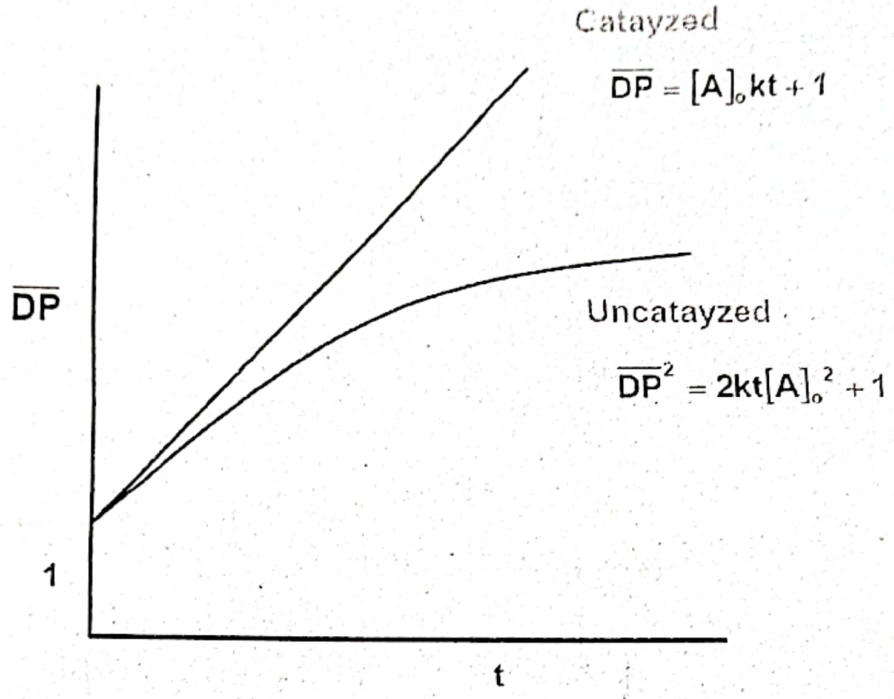
Or

$$\frac{1}{(1-p)^2} = 2kt\,[A_0]^2 + 1 \quad \text{----------} \quad (5)$$

and

$$\overline{(DP)^2} = 2kt\,[A_0]^2 + 1$$

CamScanner



Final-16

The following conditions must be fulfilled in step-growth polymerizations for higher molecular weight:

- high conversions
- high monomer purities
- high reaction yields
- stoichiometric equivalence of functional groups

CamScanner

Stoichiometric Imbalance

Despite the fact that an exact stoichiometric balance is necessary to achieve high molecular weights in most instants of step reaction polymerization, in some important application it is necessary to limit the molecular weight.

- Epoxy resins are **polyethers** that are processed to an **oligomer stage** and are subsequently converted to **network polymer** by appropriate reactions of terminal epoxyide groups.
- With **polyimides** for fiber applications, molecular weight must often be limited because **too high a viscosity** is detrimental to **extrusion of filaments through the fine holes of a spinneret**.

Three ways to *limit* M. W. in step polymerization

1. To quench the polymerization reaction by rapid cooling when the desired M.W. is attained.
2. To use an excess of one monomer.
3. To use small amounts of monofunctional reactant

CamScanner

When the nonstoichiometric amount of functional groups is used, the relation between $\overline{DP}$ and the reaction conversion can be modified by modification of the Carothers equation.

In order to do so let us introduce a factor, *r*, representing the *stoichiometric imbalance*.

·If the number of A and B functional groups present at the beginning of AA + BB polymerization is N_A^0 and N_B^0 respectively, Then stoichiometric imbalance r, is given by

$$r = \frac{N_A^0}{N_B^0}$$

By convention *r* is always less than unity.

p = reaction conversion, in this case fraction of A group that have reacted.

As reaction of each A group consumes one B group, fraction of B reacted at *p* is equal

$$pN_A^0 \text{ or } prN_B^0$$

CamScanner

The number of *unreacted* groups , N_A and N_B is given by

$$N_A = (1-p)\,N_A^0 \quad \text{and} \quad N_B = (1-pr)\,N_B^0 = (1-pr)\,\frac{N_A^0}{r}$$

At this point, the number of A and B end group is equal to $N_A + N_B$, and because there are two end groups per molecule, the number of molecular chains, N, is given by

$$N = \frac{1}{2}(N_A + N_B)$$

That is

$$N = \frac{1}{2}\left[(1-p)\,N_A^0 + (1-pr)\frac{N_A^0}{r}\right]$$

which reduces to

$$N = \frac{N_A^0}{2}\left(1 + \frac{1}{r} - 2p\right)$$

We know that one repeating unit is formed every time an A and B functional group react; therefore the total number of repeating units N_r is given by

$$N_r = \frac{1}{2}(N_A^0 + N_B^0)$$

Since, $r = \frac{N_A^0}{N_B^0}$

$$N_r = \frac{1}{2}\left(N_A^0 + \frac{N_A^0}{r}\right) = \frac{N_A^0}{2}\left(\frac{r+1}{r}\right)$$

The average degree of polymerization is equal to the number of monomeric units divided the number of chains; that is

$$\overline{DP} = \frac{N_r}{N} = \frac{(N_A^0/2)[(r+1)/r]}{N_A^0/2[1+(1/r)-2p]}$$

$$r = \frac{N_A^0}{N_B^0} < 0, \quad (N_A^0 < N_B^0)$$

TU-17

Carothers equation reduces to

$$\overline{DP} = \frac{r+1}{r+1-2pr}$$

imbalance
TU

Now if $r = 1$

$$\overline{DP} = \frac{1}{1-p}$$

balance
TU

equation only

When monomer A is completely used up in the polymerization (that is $p = 1$) the equation becomes

$$\overline{DP} = \frac{1+r}{1-r}$$

CamScanner

If monofunctional reagent is added

Imbalance factor

$$r' = \frac{N_A^0}{N_B^0 + 2\,N^0_{B'}}$$

where $N^0_{B'}$ = number of monofunctional B groups

Factor 2 : each B' molecule is equally as effective as one excess BB monomer in limiting the M.W.

CamScanner

Molecular weight distribution

In any polymerization involving difunctional monomer, each step reaction links two monomer together; thus the number of monomer units in the polymer is always one more than the functional group.

If a polymer contains x monomer units, the number of A or B groups that have reacted is x-1.

The probability that x-1 A or B groups has reacted is p^{x-1} , where p is the reaction conversion, defined previously as

$$p = \frac{N_0 - N}{N_0}$$

The probability of finding an unreacted group is

$$1 - p$$

CamScanner

The probability of finding a molecule containing x units and an unreacted A or B group is then

$$p^{x-1}(1-p)$$

If N= total number of molecule present at time t, the fraction that contains x units N_x, is given by

$$N_x = N\,p^{x-1}(1\text{-}p)$$

$\vec{DP} = \frac{N_0}{N}$

$DP = \frac{1}{1-P}$ $\therefore N = N_0(1-P)$

As $N/N_0 = 1\text{-}p$ (Carother equation), we can write the expression for N_x as

$$N_x = N_0(1\text{-}p)^2\,p^{x-1}$$

This is the number distribution functio

Where N_0 = initial number of monomers and the relationship defines number or mole fraction distribution at t.

The weight fraction w_x, of molecules containing x units is equal to the ratio of the mass of those molecules divided by the total mass; that is

$$w_x = \frac{xN_xM_0}{N_0M_0} = \frac{xN_x}{N_0}$$

Where M_0 is the mass of the repeating unit.

As, $N_x = N_0(1-p)^2 p^{x-1}$

We can write ,

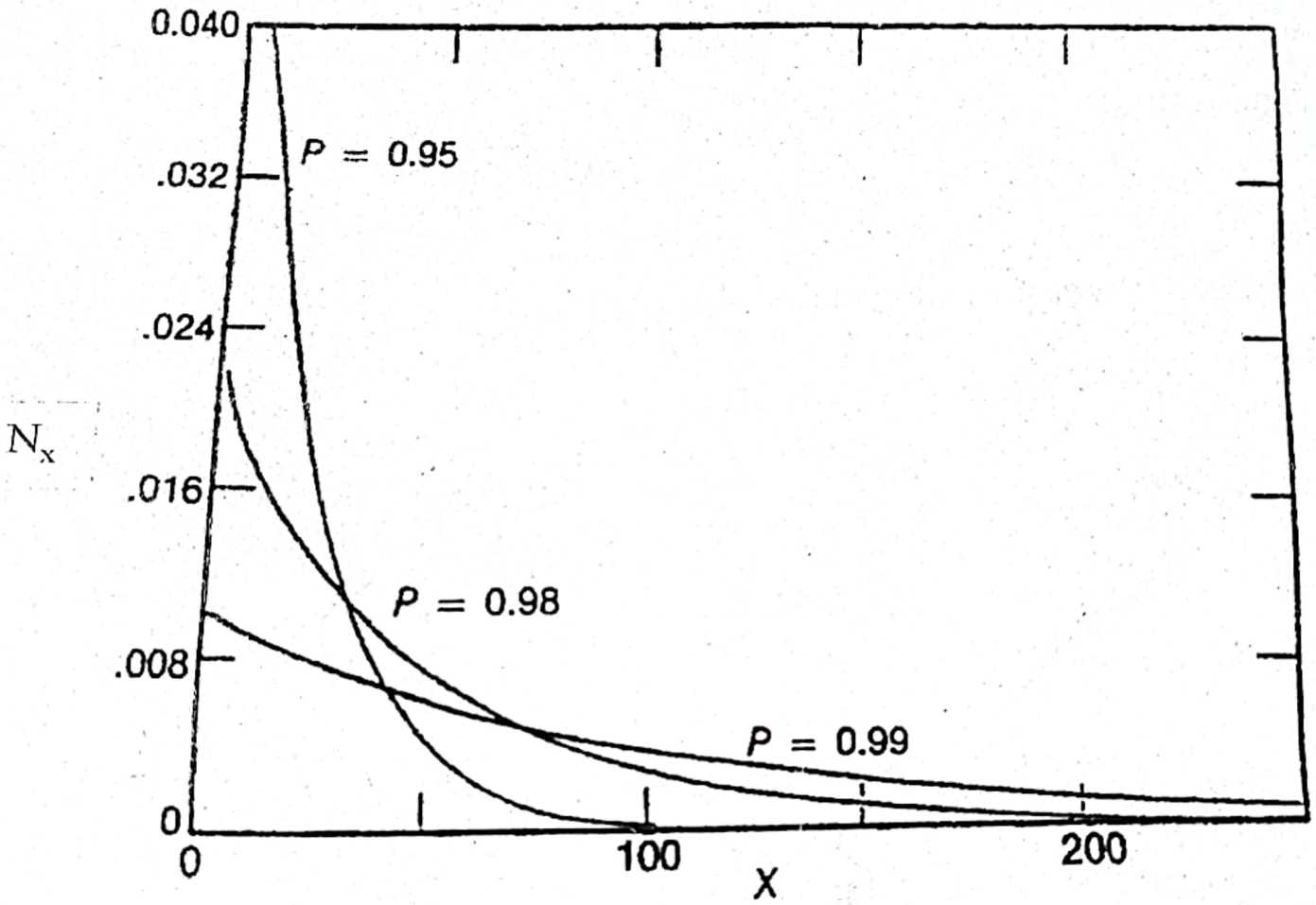
$$w_x = x(1-p)^2 p^{x-1}$$

Which is the weight fraction distribution.

CamScanner

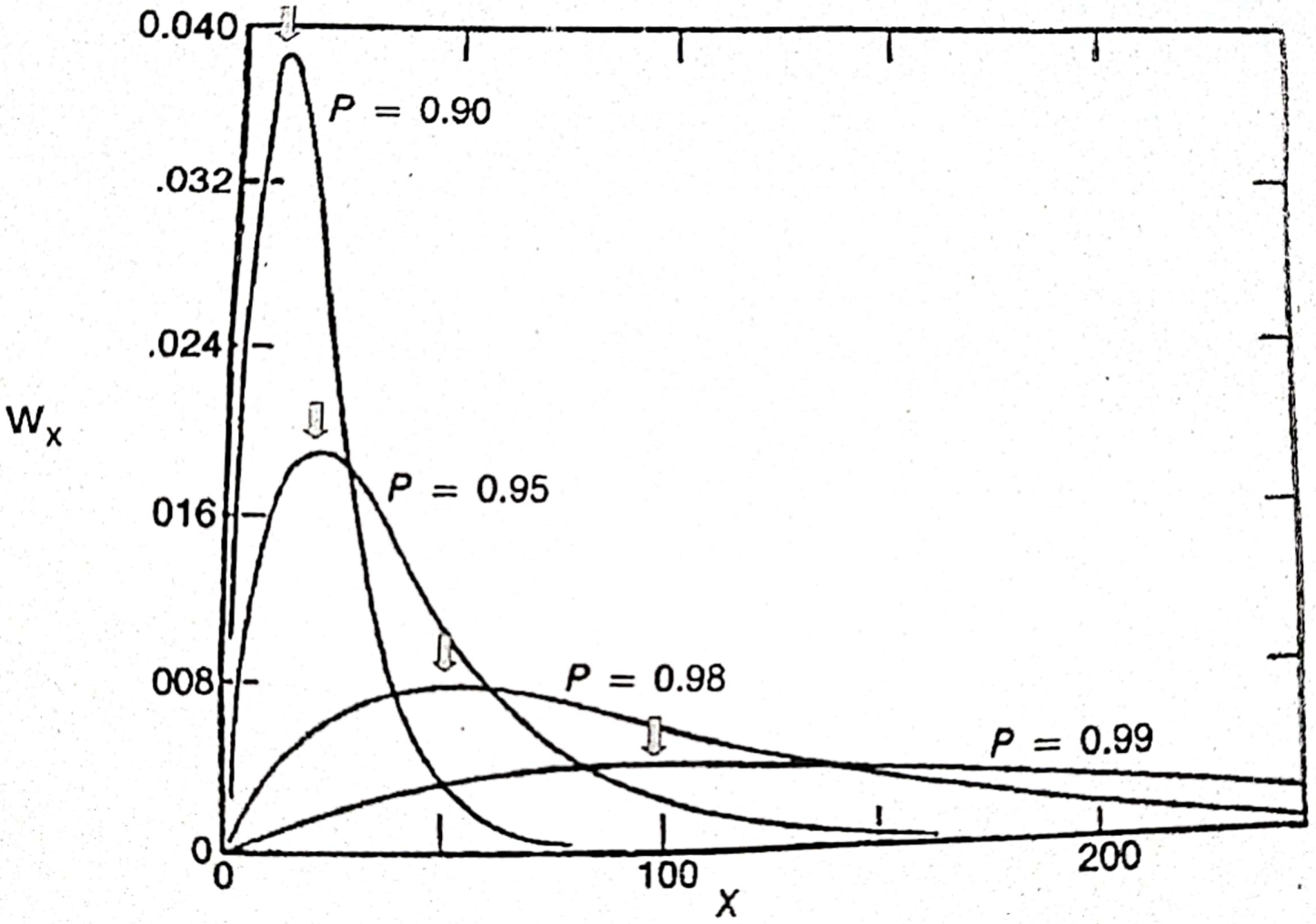
Number

Mole fraction distribution in linear step-reaction polymerization



CamScanner

! Weight fraction distribution in linear step-reaction polymerization



CamScanner